Sustainability Pioneers Corner (SP Corner)

Hostel Recycling Buy-Back Initiative: Integrated Physical-Digital Pilot & Operational Budget

Project Lead:	Fatimah Ayoola (Sustainability Pioneers)	**Target Sponsor:**	PPK (Facilities, Infrastructure & Scholarships)
Institution:	Albukhary International University (AIU)	**Pilot Scope:**	Single Hostel Focus (2 Sessions / Week)
Pilot Duration:	8 Weeks (Phase 1 Validation)	**Approved Budget Ask:**	RM 750.00 (Lean Pilot Model)

1. EXECUTIVE SUMMARY & STRATEGIC PIVOT

Following direct strategic feedback from institutional collaborators (PPK), the Sustainability Pioneers team is executing a disciplined pilot-first strategy. Rather than deploying a multi-module application across campus simultaneously, the team is proving practical operational capability through the Sustainability Pioneers Corner (SP Corner). This initiative establishes a measurable, physical-digital recycling buy-back system within campus residential hostels, validating student compliance, operational recovery rates, and backend data tracking before expanding into broader campus features.

2. PHYSICAL-DIGITAL OPERATIONAL ARCHITECTURE

The initiative tightly pairs on-site physical sorting and weighing with an internal progressive web tracker to guarantee complete data integrity, student motivation, and fraud prevention:

- **Collection Schedule:** Staffed operations run strictly twice per week (2 hours per session, totaling 4 hours weekly) at a designated hostel ground-floor station.
- **Accepted Stream Segregation:** Rigid sorting covering plastics (PET bottles, HDPE containers), packaging cardboard/paper, aluminium drink cans, and broken small appliances (kettles, irons).
- **Weight Qualification Threshold:** A minimum submission threshold of 0.5 kg qualifies for direct reward accrual. Submissions under 0.5 kg (e.g., 0.2 kg) are logged as participation check-ins, allowing students to aggregate weight across multiple drop-offs.
- **Student ID Verification:** Students log in via their institutional Student ID to view personal impact metrics. Station volunteers scan the Student ID barcode to authenticate submissions and record precise scale measurements directly into the backend.
- **Public vs. Authenticated Views:** A public dashboard displays campus-wide aggregated metrics (total kilograms collected, carbon offset estimates) accessible without sign-in, while individual submission histories and tier progression require student authentication.

3. TWO-TIER INCENTIVE & RECOGNITION STRUCTURE

To drive long-term habit formation while keeping operational costs self-funding, the reward system operates across two progressive tiers:

Tier 1: Consistency Milestone (10 Verified Submissions)

Awarded after a student completes 10 verified submissions that meet or exceed the qualifying weight threshold. Incentives consist of branded Sustainability Pioneers merchandise (custom badges, sustainable stationery, stickers, or small utility items) and a 1-week 1.5x micro-rate multiplier on eligible materials.

Tier 2: Hall of Fame & Institutional Leadership (2-Month Evaluation)

Evaluated at the conclusion of each 8-week cycle for the top active and consistent contributors. Rewards focus on high-prestige institutional recognition: featured spot on the digital campus Hall of Fame leaderboard (opt-in), formal certificates of leadership in campus sustainability endorsed by PPK/3 Zero Club, branded eco-apparel (T-shirts), and priority nomination for student leadership or community service accolades.

4. DATA GOVERNANCE, PRIVACY & PDPA COMPLIANCE

- **Data Minimization:** The digital tracking module records only essential operational parameters: Student ID, timestamp, material category, and net scale weight. No financial account numbers or personal passwords are stored.
- **Public Anonymity:** Public leaderboards display optional display names or masked student IDs (e.g., AIU***821). Full Student IDs are never exposed on unauthenticated screens.
- **Role-Based Volunteer Access:** Station volunteers hold restricted input permissions to log verified weights. Administrative rights (rate adjustments, audit exports, log overrides) remain restricted to the Project Lead.

5. LEAN PILOT BUDGET & FINANCIAL ALLOCATION

Item Category	Description & Justification	Allocation	Funding Model
Physical Equipment	1x calibrated 40kg heavy-duty digital hanging/ platform scale & safety gloves	**RM 150.00**	PPK Seed Grant
Sorting & Storage	3x heavy-duty stackable segregated collection bins (plastics, cans, paper)	**RM 120.00**	PPK Seed Grant
Digital Infrastructure	Cloud hosting & database backend for pilot drop logger (3-month runtime)	**RM 180.00**	PPK / Club Budget
Tier 1 Student Rewards	Branded merchandise (pins, custom stickers, sustainable stationery) for milestone earners	**RM 150.00**	Recycler Resale Float
Operational Float & Buffer	Emergency petty cash for scale calibration, bin liners, and initial payout buffer	**RM 150.00**	Revolving Float
Total Pilot Capital Ask	**Complete 8-week single-hostel verification operational budget**	**RM 750.00**	**Self-Sustaining Target**

6. 8-WEEK PILOT EXECUTION TIMELINE

- **Weeks 1–2 (Station Setup & Recycler Onboarding):** Secure PPK storage clearance in pilot hostel; procure digital scale and sorting bins; execute bulk purchase agreement with local Alor Setar recycling aggregator.
- **Weeks 3–4 (Drop-Off Launch & Volunteer Calibration):** Commence twice-weekly collection sessions; test volunteer data entry via mobile web scanner; calibrate weight accuracy.
- **Weeks 5–6 (Mid-Pilot Audit & Tier 1 Milestones):** First bulk sale of collected materials to recycler; evaluate student compliance and payout margins; issue first batch of Tier 1 milestone incentives.
- **Weeks 7–8 (Final Evaluation & PPK Presentation):** Complete pilot reconciliation; publish net waste diversion statistics; present comprehensive impact report to PPK to unlock Phase 2 multi-hostel expansion.